

OIL INDIA LIMITED - eRTMAC OPERATIONS

OFFICIAL WELL COMPLETION REPORT (WCR) & DRILLING HISTORY
WELL IDENTIFIER: 15/9-23

1. Well Location & Technical Header

Operator: Oil India Limited / Joint North Sea	Well Name: 15/9-23
UTM Easting (X): 433919.0 m	UTM Northing (Y): 6459992.0 m
Latitude: 58.276005° N	Longitude: 1.873338° E
Total Depth (MD): 3212.6 m	Spud Depth: 1518.3 m
Drilling Rig: eRTMAC Semi-Submersible Rig-07	Status: Plugged & Abandoned (P&A;) / Offset Reference

2. Stratigraphic Progression & Formation Tops

Formation / Stratum Name	Top Depth MD (m)	Lithology Characterization	Drillability / Risk Profile
Skade Fm.	1518.3 m	Interbedded Sand / Shale	Loss / Depletion
Frigg Fm.	1607.5 m	Interbedded Sand / Shale	Loss / Depletion
Balder Fm.	1696.6 m	Interbedded Sand / Shale	Loss / Depletion
Sele Fm.	1785.8 m	Shale / Claystone with interbedded siltstone	Tight Hole / Swelling
Lista Fm.	1875.0 m	Interbedded Sand / Shale	Loss / Depletion
Ty Fm.	1964.2 m	Porous Turbidite Sandstone (Hydrocarbon Bearing)	Loss / Depletion
Ekofisk Fm.	2053.3 m	Chalk / Hard Calcareous Limestone	Fractured / Losses
Tor Fm.	2142.5 m	Chalk / Hard Calcareous Limestone	Fractured / Losses
Hod Fm.	2231.7 m	Interbedded Sand / Shale	Loss / Depletion
Blodoeks Fm.	2320.9 m	Interbedded Sand / Shale	Loss / Depletion
Hidra Fm.	2410.0 m	Interbedded Sand / Shale	Loss / Depletion
Roedby Fm.	2499.2 m	Interbedded Sand / Shale	Loss / Depletion
Sola Fm.	2588.4 m	Interbedded Sand / Shale	Loss / Depletion
Tuxen Fm.	2677.6 m	Interbedded Sand / Shale	Loss / Depletion
Aasgard Fm.	2766.7 m	Interbedded Sand / Shale	Loss / Depletion
Draupne Fm.	2855.9 m	Interbedded Sand / Shale	Loss / Depletion
Heather Fm.	2945.1 m	Interbedded Sand / Shale	Loss / Depletion
Sleipner Fm.	3034.3 m	Interbedded Sand / Shale	Loss / Depletion
Skagerrak Fm.	3123.4 m	Interbedded Sand / Shale	Loss / Depletion

3. Casing Design & Mud Program Summary

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36 in Hole	30 in Conductor	350 m	Spud Mud / Seawater	1.05 SG	FIT 1.25 SG
26 in Hole	20 in Surface	1,050 m	KCl / Polymer	1.12 SG	LOT 1.45 SG
17-1/2 in Hole	13-3/8 in Interm.	2,200 m	Low Solids Non-Dispersed	1.24 SG	LOT 1.62 SG
12-1/4 in Hole	9-5/8 in Prod.	2913 m	Oil Based / Synthetic Mud	1.32 SG	LOT 1.78 SG
8-1/2 in Hole	7 in Liner	3213 m	Synthetic Oil Based (SOBM)	1.38 SG	LOT 1.88 SG

4. Historical Operational Incidents, NPT & Mitigations (Well: 15/9-23)

This section constitutes the institutional memory log extracted by the NWIS knowledge engine. All alerts generated during active drilling cite these exact records.

INCIDENT #1: Lost Circulation / Mud Loss	Severity: HIGH	NPT Hours: 18.5 hrs
Depth Interval: 1680.0 m - 1715.0 m	Formation: HORDALAND GP	Status: Resolved on Site
Event Occurrence Narrative: Encountered sudden severe lost circulation while drilling 12-1/4 inch hole at 1695m in micro-fractured Hordaland shale. Flow-out sensor dropped to 15% with total mud loss rate of 65 bbl/hr. Total fluid lost: 340 bbls.		
Mitigation Action Applied: Pulled bit off bottom into 13-3/8 inch casing shoe. Mixed and pumped 50 bbl high-fluid-loss LCM pill containing 25 ppb medium walnut shell and 15 ppb fibrous mica. Waited 4 hours for pill to set. Resumed drilling with reduced pump rate (550 gpm) and maintained mud weight at 1.18 SG. Full returns restored.		
Offset Drilling Recommendation & Institutional Lesson: When drilling through upper Hordaland fractured intervals within 1650-1750m, pre-treat active mud system with fine carbonate and limit ECD to under 1.22 SG.		

INCIDENT #2: Stuck Pipe Incident	Severity: CRITICAL	NPT Hours: 26.0 hrs
Depth Interval: 2420.0 m - 2445.0 m	Formation: TY FM	Status: Resolved on Site
Event Occurrence Narrative: String became differentially stuck during connection at 2432m in permeable Ty sandstone with 85 psi overbalance. Unable to rotate or pull string with 90 klbs overpull.		
Mitigation Action Applied: Spotted 45 bbls organic pipe-freeing soak pill across Ty sand interval. Allowed 3.5 hours soak time while applying intermittent torque and 110 klbs upward jar impacts. String freed on 14th jar stroke. Circulated hole clean and increased mud yield point to 22 lb/100ft2.		
Offset Drilling Recommendation & Institutional Lesson: Avoid static drill string conditions exceeding 3 minutes in Ty formation. Keep pipe rotating during survey/connection procedures.		